Sometimes the integrand needs to be rewritten before using trig substitution. We now go through some examples.

1. Evaluate $\int \frac{dx}{x^4\sqrt{9x^2-1}}$. (*Hint*: First rewrite $\sqrt{9x^2-1}$ so that x^2 has coefficient 1.)

2. Evaluate $\int_0^{3\sqrt{3}/2} \frac{x^3}{(4x^2+9)^{3/2}}dx$. (*Hint*: Note that $(4x^2+9)^{3/2} = (\sqrt{4x^2+9})^3$.)

3. Evaluate $\int \frac{x}{\sqrt{3-2x-x^2}}dx$. (*Hint*: First complete the square under the radical.)

4. Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.